Predict the Output

Python Programming Exercises

Instructions:

- For each question, carefully read the Python code
- Predict what will be displayed when the program runs
- Write your answer in the space provided
- Consider operator precedence, data types, and control flow
- If there are multiple lines of output, write each on a separate line
- If the program asks for input, assume the input values are provided as mentioned

Easy Level Questions (1-10)

These questions test basic understanding of print(), variables, and simple operations.

Question 1:

```
print("Hello World")
print("Python is fun!")
```

Your Answer:

Question 2:

```
x = 5
y = 3
print(x + y)
```

Your Answer:

Question 3:

```
name = "Alice"
age = 25
print("Name:", name)
print("Age:", age)
```

Your Answer:

Question 4:

```
a = 10
b = 4
print(a - b)
print(a * b)
```

Your Answer:

Question 5:

```
score = 85
if score > 80:
    print("Good job!")
```

Your Answer:

Question 6:

```
temperature = 15
if temperature > 20:
    print("Warm day")
else:
    print("Cool day")
```

Your Answer:

Question 7:

```
x = 7
y = 7
print(x == y)
print(x != y)
```

Your Answer:

Question 8:

```
num = 12
print(num / 3)
print(num // 3)
```

Your Answer:

Question 9:

```
value = 50
if value >= 50:
    print("Pass")
```

Your Answer:

Question 10:

```
a = "Hello"
b = "World"
print(a + " " + b)
```

Your Answer:

Medium Level Questions (11-20)

These questions involve multiple concepts and require careful analysis of operator precedence and control flow.

Question 11:

```
x = 8
y = 3
print(x ** y)
print(x % y)
```

Your Answer:

Question 12:

```
age = 17
if age >= 18:
    print("Adult")
else:
    print("Minor")
print("Processing complete")
```

Your Answer:

Question 13:

```
a = 5
b = 10
c = a + b * 2
print(c)
print(a < b and b < 20)
```

Your Answer:

Question 14:

```
score = 75
if score >= 90:
    print("A grade")
elif score >= 80:
    print("B grade")
elif score >= 70:
    print("C grade")
else:
    print("Need improvement")
```

Your Answer:

Question 15:

```
x = 12
y = 5
result = x > y or y > 10
print(result)
print(not result)
```

Your Answer:

Question 16:

```
num1 = 15
num2 = 20
if num1 > 10 and num2 > 10:
    sum_result = num1 + num2
    print("Sum is:", sum_result)
```

Your Answer:

Question 17:

```
value = 0
if value:
    print("True")
else:
    print("False")
print(value == False)
```

Your Answer:

Question 18:

```
a = 6
b = 4
c = 2
result = a + b * c - 1
print(result)
print(result > a + b)
```

Your Answer:

Question 19:

```
temperature = 22
humidity = 60
if temperature > 20 and humidity < 70:
    print("Comfortable weather")
elif temperature > 20:
    print("Warm but humid")
else:
    print("Cool weather")
```

Your Answer:

Question 20:

```
x = 3
y = 7
z = x * 2 + y // 2
print(z)
if z > 8:
    print("Greater than 8")
    z = z - 5
    print("New value:", z)
```

Your Answer:

Hard Level Questions (21-30)

These questions test complex logic, edge cases, and advanced understanding of Python concepts.

Question 21:

```
a = 4
b = 6
c = 8
if a + b > c and a + c > b and b + c > a:
    print("Valid triangle")
else:
    print("Invalid triangle")
print("Sum of all sides:", a + b + c)
```

Your Answer:

Question 22:

```
x = 2
y = 3
z = 4
result = x ** y + z * (x + y) - z // x
print(result)
if result % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Your Answer:

Question 23:

```
age = 25
income = 30000
if age >= 21 and income >= 25000:
    if age < 30:
        print("Young professional")
    else:
        print("Experienced professional")
elif age >= 21:
    print("Young adult")
else:
    print("Student")
```

Your Answer:

Question 24:

```
p = True
q = False
r = True
result1 = p and q or r
result2 = p and (q or r)
print(result1)
print(result2)
print(result1 == result2)
```

Your Answer:

Question 25:

```
num = 127
if num > 100:
    if num % 2 == 1:
        if num % 3 == 1:
            print("Special number")
        else:
            print("Odd but not special")
    else:
        print("Even and large")
else:
    print("Small number")
```

Your Answer:

Question 26:

```
a = 5
b = 0
c = 10
if a and b and c:
    print("All true")
elif a and c:
    print("A and C are true")
elif a or b or c:
    print("At least one is true")
else:
    print("All false")
```

Your Answer:

Question 27:

```
x = 3
y = 5
z = 2
max_val = x
if y > max_val:
    max_val = y
if z > max_val:
    max_val = z
print("Maximum value:", max_val)
print("Is it greater than average?", max_val > (x + y + z) / 3)
```

Your Answer:

Question 28:

```
score1 = 85
score2 = 92
score3 = 78
average = (score1 + score2 + score3) / 3
print("Average:", average)
if average >= 90:
    grade = "A"
elif average >= 80:
    grade = "B"
elif average >= 70:
    grade = "C"
else:
    grade = "F"
print("Grade:", grade)
print("Pass" if grade != "F" else "Fail")
```

Your Answer:

Question 29:

```
year = 2024
if year % 4 == 0:
    if year % 100 == 0:
        if year % 400 == 0:
            print("Leap year")
        else:
            print("Not a leap year")
    else:
        print("Leap year")
else:
    print("Not a leap year")
print("Year processed:", year)
```

Your Answer:

Question 30:

```
a = 7
b = 3
c = 5
# Complex expression evaluation
result = a > b and b < c or not(a == c) and (a + b) % c == 0
print("Result:", result)
if result:
    final_value = a * b + c
else:
    final_value = a + b - c
print("Final value:", final_value)
print("Type check:", type(final_value).__name__)
```

Your Answer: